Markscheme

Mathematics: analysis and approaches
Practice question bank

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158. (a)

Given $z = 2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$.

Use De Moivre's theorem to find z^5 in the form $r(\cos\theta + i\sin\theta)$, stating r and θ.

$$r = 2^5 = 32$$ *(M1)*

$$\theta = 5 \times \frac{\pi}{6} = \frac{5\pi}{6}$$ *(M1)*

$$\approx 2.618$$ **A1**

[4 marks]